

Aditya Awathare

AI/ML Engineer Intern | Agentic AI & Backend Developer | Nagpur

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Professional Summary

B.Tech Computer Engineering graduate with production-grade experience in Python, REST APIs, AWS, and backend systems. Actively building expertise in Agentic AI, OpenAI API integration, and multi-step tool-using workflows. Strong foundation in API integration, OOP, and cloud deployment. No Active Backlogs | Immediate Joiner.

Technical Skills

AI & LLM: Python, Agentic AI, RAG (Qdrant, Chroma), LangGraph/LangChain, Groq/OpenAI API, MCP, Prompt Engineering, Embeddings, TensorRT/OpenVINO

Backend: FastAPI, Django, REST APIs, Microservices, Redis, WebSockets, Webhooks, TypeScript, OOP

Cloud & Infrastructure: AWS (EC2, S3, IAM, Route 53, CloudFront, Lambda), Docker, Linux

Tools: Git/GitHub, CI/CD

Experience

Yugavist InfraDesign, Nagpur - AI/ML Intern Apr 2026 - Present

Python, FastAPI, RAG, Groq/OpenAI, Qdrant, YOLO, AWS

- Building internal RAG assistant over tender documents and project drawings - ingested PDFs with chunked embeddings (Qdrant/Chroma) and keyword/synonym fallback, reducing manual lookup for estimation team.
- Prototyping site-progress vision pipeline with YOLO for asset detection from site photos; exposing inference via FastAPI for internal dashboard integration.
- Automating lead capture and FAQ workflows with ReAct-style tool-using agent (JSON tool calls to execution to natural reply) and evaluation harness for reliability; pattern later open-sourced as VoiceOps.

Projects

VoiceOps - AI Voice Receptionist (Agentic AI) Aug 2026

Python, FastAPI, Groq, RAG, MCP, Twilio

- Wanted an AI receptionist that could take action (book a slot), not just answer questions - built a ReAct-style agent loop where the LLM emits a JSON tool call, the tool executes, and a second LLM call turns the result into a natural spoken reply.
- Needed the demo to stay reliable without live keys, so added a Groq to OpenAI to local mock fallback chain and hash-stable demo calendar slots.
- Answered business FAQs with a RAG pipeline over the salon knowledge base, falling back to keyword/synonym matching when retrieval was uncertain.
- Exposed the same five tools (availability, booking, FAQ, lead logging, human handoff) over both a FastAPI voice/chat API and an MCP server, drivable from a browser mic, a Twilio call, or an MCP client.
- Wrote an 8-scenario evaluation suite (incl. Hinglish input); all 8 currently pass at ~1.1s average latency.

[View repository ->](#)

Open Source Contribution - Ultralytics YOLO (GitHub) May 2026

Python, TensorRT, OpenVINO, Model Export Pipelines, Pytest

- Noticed YOLO INT8 export pipeline reused the same batch size for inference and calibration, limiting flexibility for memory-constrained TensorRT/OpenVINO calibration runs - proposed and implemented an optional `calibration_batch` parameter to decouple the two.
- Extended the exporter, config validation, and calibration dataloader to support the new parameter with a safe fallback to the existing batch size, and added regression tests covering both export formats.
- Iterated through two rounds of maintainer code review - replacing an assert with proper input validation, removing an unrelated config default - to align with project conventions.
- Reviewed and approved by maintainers; PR was later closed without merge.

[View pull request \(closed\) ->](#)

Automated Deployment Pipeline - GitHub to Live URL (AWS S3) Mar 2026

Node.js, TypeScript, AWS S3, Redis, WebSockets, React

- Wanted to understand how platforms like Vercel deploy a frontend from a GitHub URL, so built a 4-service pipeline: an upload service that clones the repo and pushes source to S3, a queue-based build worker, a request handler that serves built files by host, and a React UI to trigger/monitor deploys.
- Used Redis as a build queue to decouple upload from build/deploy, and streamed live build logs to the frontend over WebSockets instead of polling.
- Solved the is it actually live problem by tracking per-deployment status in Redis and scoping the S3 bucket policy to that deployment path, so a URL only goes live once the build succeeds.

[View repository ->](#)

Cloud File Storage App - Django + AWS S3 Feb 2026

Python, Django, boto3, AWS S3

- Built a Django app to manage S3 files end-to-end - upload, list with size/last-modified, secure download via presigned URLs, and delete - instead of exposing the bucket directly.
- Large files were failing on single-request uploads, so implemented browser-side chunked multipart upload with backend-signed part URLs, a live progress bar with speed/ETA, and completion/abort handling.

[View repository ->](#)

Education

Smt. Radhikatai Pandav College of Engineering, Nagpur 2021 - 2025

B.Tech Computer Engineering | CGPA: 7.0

Tip Top Convent, Nagpur 2020 - 2021

Higher Secondary Education

Certifications

Oracle Certified Foundations Associate - Agentic AI

HackerRank (2024-2026): SQL Advanced | Python Programming | JavaScript Fundamentals

Additional Information

Notice Period: Immediate | Languages: English (Fluent), Hindi (Native), Marathi (Native)